How the Internet Works: DNS, IP & URLs

Follow a web request from the address bar to the rendered page.

1. What happens when you type a URL?

The internet is a global network of networks. Your browser, device, router, internet provider, and the website's servers cooperate to exchange small units of data. A webpage may appear in a second, but several steps can happen behind the scenes.

A URL (Uniform Resource Locator) identifies a resource. For example, it can contain a scheme such as https, a domain name such as example.com, and a path such as /articles. The browser interprets the URL to determine how and where to request the resource.

Step 1 — Browser checks

The browser may reuse cached files or a previously resolved domain address. Caching saves time and network traffic. If a suitable copy is available and still valid, the browser may not need to fetch every item again.

2. DNS: names to addresses

People remember domain names more easily than numeric IP addresses. The Domain Name System (DNS) helps translate a domain name into an IP address. The browser or operating system may check its cache first; otherwise, a DNS resolver can look up the answer through the DNS system.

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DNS in plain language

Think of DNS as a directory service: you ask for a domain name and receive information that helps your device locate the server. DNS does not normally deliver the webpage itself; it helps find where to send the request. A domain can have multiple addresses, and the result can vary with caching and network configuration.

3. Making a connection

After the address is known, the browser establishes a connection to the server. Many HTTPS connections use TCP plus TLS; HTTP/3 instead uses QUIC over UDP. The exact sequence depends on the protocol, whether a connection can be reused, and whether the browser has already established a secure session.

TCP and TLS

TCP provides reliable, ordered delivery of a byte stream between endpoints. **TLS** protects data in transit through encryption and integrity checks, and it helps the browser verify the server's identity. HTTPS is HTTP carried over a secure connection; it does not automatically mean the website's content is trustworthy.

4. Sending the HTTP request

HTTP defines how clients and servers exchange requests and responses. A request can include a method, a path, headers, and sometimes a body. GET is commonly used to retrieve a resource. Headers can describe formats the browser accepts, caching rules, language preferences, and other context.

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5. The server responds

The server processes the request and sends an HTTP response. A response includes a status code, headers, and usually a body. Status code 200 commonly indicates success; 301 or 302 indicates a redirect; 404 means the requested resource was not found; and 500 indicates a server-side error.

A webpage is often made of many resources, not one file. After receiving HTML, the browser may request stylesheets, scripts, images, fonts, and data from additional endpoints. Some resources may come from a content delivery network (CDN) closer to the user.

6. Browser rendering

The browser parses HTML to build the page's document structure. It processes CSS to determine layout and appearance, runs JavaScript when needed, calculates positions and sizes, and paints pixels to the screen. These steps can overlap with downloading resources, so the page may become visible before every asset finishes loading.

Key terms

Term	Meaning
IP address	Address used to route network traffic to a device or service.
DNS	System that helps map domain names to IP addresses.
HTTP	Protocol for web requests and responses.
TLS	Security protocol protecting data in transit.
Cache	Stored copy of data that may avoid a repeat fetch.

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The full sequence at a glance

Type URL → browser/cache checks → DNS lookup if needed → connection and security setup → HTTP request → server response → browser parses and renders. This is a helpful mental model, although real browsers can reuse connections, load several resources in parallel, and render content while more data is arriving.

Common misunderstandings

• DNS finds an address; it does not usually return the page itself.
• An IP address identifies a network endpoint; it is not the same as a domain name.
• HTTPS protects data in transit, but cannot guarantee that a site is honest.
• A 404 response means the resource was not found; it does not necessarily mean the entire server is down.
• Rendering can begin before all images and scripts finish loading.

Quick self-check

1. What problem does DNS solve?
2. What is the role of an IP address?
3. What is one difference between TCP and TLS?
4. What do status codes 200 and 404 usually mean?
5. Name two things the browser uses to render a page.

Answer key

1. It helps translate domain names into address information. **2.** It helps route traffic. **3.** TCP transports data reliably; TLS protects it cryptographically. **4.** Success and resource not found. **5.** HTML and CSS (JavaScript may also be involved).